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DoF-Gaussian: Controllable Depth-of-Field for 3D Gaussian Splatting

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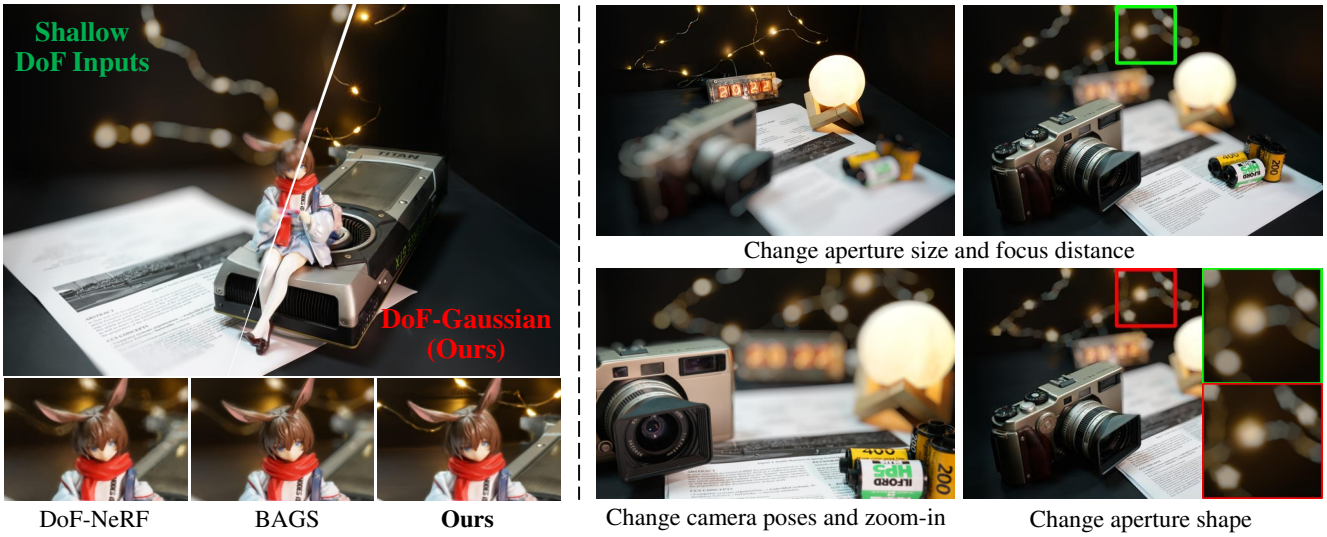


Figure 1. Given a set of multi-view input images with shallow DoF, DoF-Gaussian can reconstruct a 3D-GS representation of a sharp scene. Thanks to our lens-based design, we can also achieve controllable DoF effects for a variety of applications. The example input images are taken from [55] for illustration purposes. **(Zoom-in for best view)**

Abstract

Recent advances in 3D Gaussian Splatting (3D-GS) have shown remarkable success in representing 3D scenes and generating high-quality, novel views in real-time. However, 3D-GS and its variants assume that input images are captured based on pinhole imaging and are fully in focus. This assumption limits their applicability, as real-world images often feature shallow depth-of-field (DoF). In this paper, we introduce DoF-Gaussian, a controllable depth-of-field method for 3D-GS. We develop a lens-based imaging model based on geometric optics principles to control DoF effects. To ensure accurate scene geometry, we incorporate depth priors adjusted per scene, and we apply defocus-to-focus adaptation to minimize the gap in the circle of confusion. We also introduce a synthetic dataset to assess refocusing capabilities and the model's ability to learn precise lens parameters. Our framework is customizable and supports various interactive applications. Extensive experiments confirm the effectiveness of our method. Our project is available at https://dof-gaussian.github.io/.

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1. Introduction

Depth-of-field (DoF) refers to the distance between the closest and farthest objects in a photo that appears acceptably sharp. In practice, photographers can control the DoF by adjusting the camera's aperture size or focus distance to capture images with either a wide DoF (all-in-focus) or shallow DoF (defocused). The shallow DoF effect is an important technique in photography, as it draws the viewers' attention to the focal region by blurring the surrounding areas. Meanwhile, novel view synthesis aims to create realistic images from novel viewpoints based on a set of source images. However, novel view synthesis typically requires all-in-focus input images and lacks the capability to render varied DoF effects, limiting its applications. In this work, we aim to render novel views with controllable DoF, adding cinematic quality to the results.

Images captured from the real world sometimes have shallow DoF. Specifically, points of light that do not lie on

the focal plane are projected onto the sensor plane as blurred circles, referred to as the circle of confusion (CoC), and this introduces bokeh blur to the captured image. Most novel view synthesis methods experience performance degradation when processing shallow DoF input images. To address this issue, DoF-NeRF [55] and LensNeRF [17] introduce lens-based camera models in volume rendering to explicitly enable controllable DoF effects. Meanwhile, methods like Deblur-NeRF [26], DP-NeRF [19], and PDRF [33] employ sparse kernels to model defocus blur. However, implicit rendering approaches face significant challenges in training and rendering efficiency. In the realm of 3D Gaussian Splatting (3D-GS) [16], methods such as BAGS [34] and Deblurring 3DGS [18] propose blur estimation networks to model the blur kernels or scaling factors for defocus deblurring, but they cannot accommodate controllable DoF.

In this paper, we present DoF-Gaussian, an efficient framework for controllable DoF in 3D-GS, addressing the limitations of existing methods in handling shallow DoF inputs. This problem is non-trivial because 3D-GS and its variants [11, 23, 54, 57, 58] are typically based on a pinhole camera model that assumes all-in-focus inputs, meaning both foreground and background appear clear. When input images contain bokeh blur, it becomes challenging to accurately construct scene geometry and render depth maps. However, rendering images with different DoFs relies heavily on precise depth estimation. Additionally, our 3D-GS approach assumes an idealized CoC, which inevitably differs from the CoC seen in real photographs, posing further challenges for accurate defocus deblurring. Furthermore, existing datasets in this field are primarily borrowed from deblurring applications, limiting our evaluation to the model's defocus deblurring capability and overlooking other aspects of controllable DoF effects.

To address these challenges, we present a new controllable DoF 3D-GS method with the following **contributions**: ***First***, we employ a lens-based model instead of a pinhole-based one. We make the lens optical parameters, such as aperture size and focus distance, learnable to enable control over the depth of field. ***Second***, we introduce per-scene adjustments for depth priors to minimize errors in scene geometry and depth relationships, ensuring accurate depth-of-field control. ***Third***, we propose a defocus-to-focus adaptation strategy that focuses on learning the focal region after the learnable lens parameters have converged, compensating for differences in CoC. ***Finally***, we present a synthetic dataset designed to comprehensively evaluate the model, including its refocusing capabilities and its ability to learn accurate aperture size and focus distance.

Benefiting from the lens-based imaging model, our approach enables controlled depth of field, allowing it to handle not only shallow DoF image inputs but also unlock a range of interactive and engaging applications, as depicted in Figure 1. For instance, users can render novel view images with varied DoF, refocus on custom datasets, and even adjust the shape of CoC, *e.g.*, from circular to pentagonal or hexagonal CoC . Moreover, users can dynamically change the DoF while moving the camera or zooming, rendering videos with cinematic effects. Extensive experiments demonstrate that our method outperforms state-of-the-art depth-of-field and defocus deblurring methods.

2. Related Work

Depth-of-Field Rendering. Rendering DoF effects from a single all-in-focus image has been widely explored in previous work. Bokeh blur occurs when light is projected onto the image plane as a circular region, rather than as a point. The size of CoC is affected by the diameter of the aperture (aperture size) and the distance from the camera to the focal plane (focus distance). Photos captured with a small aperture usually present a wide DoF, *i.e.*, all objects appear sharp. In contrast, as the aperture diameter increases, objects near the focal plane remain sharp, while those farther away become blurred with a larger CoC. Physically based methods [1, 20] rely on 3D scene geometry information and are time-consuming. Some methods use neural networks [8, 12–14, 38] trained end-to-end to obtain shallow DoF images. The DoF effects in some studies [6, 25, 35–37, 43, 45, 48, 56, 61] are controllable but usually require an extra disparity map. Wang *et al*. [46] combines neural fields with an expressive camera model to achieve all-in-focus reconstruction from an image stack. In this work, we show the connection between DoF rendering and novel view synthesis using a lens-based camera model.

Image Deblurring. Blur can generally be categorized into two main types: camera motion blur and defocus blur. Previous studies [7, 18, 24, 26, 33, 34, 44] attempt to address this issue. Our task of rendering all-in-focus and sharp novel view images from shallow DoF inputs can be considered as a form of defocus deblurring. By modeling the underlying physical principles, we can achieve defocus deblurring by simulating the circle-of-confusion. Furthermore, our method allows for adjustments to lens parameters, enabling applications such as refocusing—capabilities that deblurring approaches like BAGS [34] cannot achieve. Existing datasets in the 3D DoF field are often borrowed from the deblurring domain [26] or designed solely to assess defocus deblurring ability [55]. To support a wider range of applications and validate the accuracy of the learned lens model, we introduce a synthetic dataset for more comprehensive evaluations.

Novel View Synthesis. Novel view synthesis allows the rendering of unseen camera perspectives from 2D images and their corresponding camera poses. Recent advancements in synthesis can largely be attributed to Neural Radiance Field [28] and 3D Gaussian Splatting [16]. However,

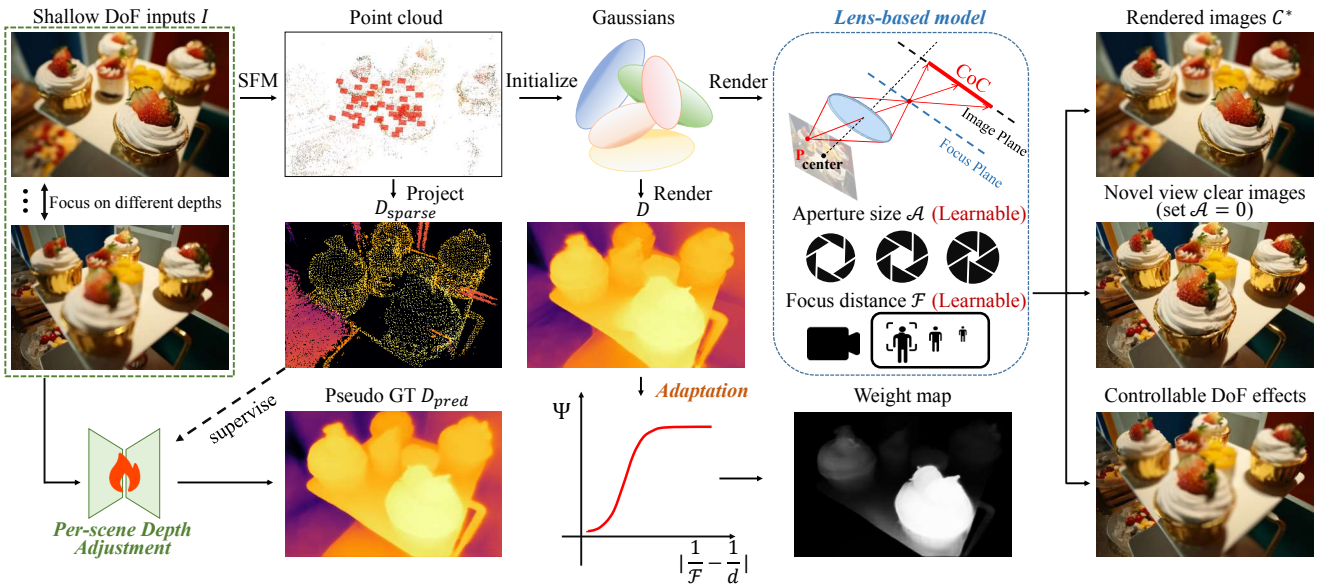


Figure 2. **Overview of DoF-Gaussian.** Given input images I with shallow DoF , we first apply SfM from COLMAP to obtain sparse depth D_{sparse}, which is used to train a depth network to derive per-scene depth priors D_{pred}. We then employ D_{pred} to regularize the Gaussians rendered depth map D. Next, by developing a lens imaging model, we can render defocused images C^* to simulate input images. To minimize the discrepancy in CoC, we propose an adaptation using the weight map. Finally, we can render fully clear images for novel view synthesis and achieve various effects by our controllable DoF framework.

real-world images are often not clean and well-calibrated. Previous work [4, 27, 29, 31, 42, 44, 49, 54, 62] attempts to cope with non-ideal conditions, such as low light, camera motion, bokeh effects, and various types of image degradation. To recover a sharp scene from images with bokeh blur, methods like Deblur-NeRF [26], DP-NeRF [19], and PDRF [33] model defocus blur using sparse kernels. NeRFocus [50] and LensNeRF [17] achieve defocus effects by incorporating a lens-based camera model in volume rendering. While DoF-NeRF [55] is similar to our approach, it is limited by the training and rendering efficiency and quality of NeRF. BAGS [34] and Deblurring 3DGS [18] use blur estimation networks to model blur kernels or scaling factors but cannot achieve controllable DoF effects due to the absence of a lens imaging model. Other studies, such as [15, 47], focus on HDR tasks and consider depth-of-field simultaneously. RGS [7] addresses defocus blur by introducing an offset to the 2D covariance matrices of Gaussians. [30] introduces a ray tracing rendering algorithm for particle-based representations to enable many advanced techniques, such as the shallow depth-of-field effect. A concurrent work [51] also explores adjustable depth of field. Our method differs in its controllable DoF mechanism, with the introduction of depth priors to ensure accurate scene depth. In addition, our new defocus-to-focus adaptation strategy eliminates the need for the focus localization network proposed in their work. Besides, we introduce a synthetic dataset for comprehensive evaluation.

3. Preliminary

3D Gaussian Splatting represents a 3D scene as a mixture of anisotropic 3D Gaussians, where each Gaussian is characterized by a 3D covariance matrix Σ and mean μ:

$$G(X) = e^{-\frac{1}{2}(X-\mu)^{\mathsf{T}}\Sigma^{-1}(X-\mu)} \tag{1}$$

The covariance matrix Σ holds physical meaning only when it is positive semi-definite. Therefore, to enable effective optimization via gradient descent, Σ is decomposed into a scaling matrix S and a rotation matrix R, as:

$$\Sigma = RSS^{\mathsf{T}}R^{\mathsf{T}}. \tag{2}$$

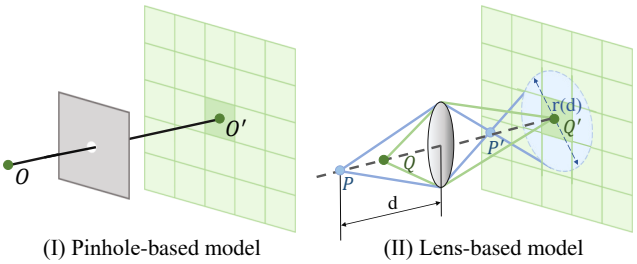
To splat Gaussians from 3D space to a 2D plane, the view transformation matrix W and the Jacobian matrix J, which represents the affine approximation of the projective transformation, are utilized to obtain the covariance matrix Σ' in 2D space, as:

$$\Sigma' = JW\Sigma W^{\mathsf{T}}J^{\mathsf{T}}. \tag{3}$$

Subsequently, a point-based alpha-blending rendering can be performed to determine the color of each pixel:

$$C = \sum_i c_i \alpha_i \prod_{j=1}^{i-1}(1-\alpha_i), \tag{4}$$

where c_i represents the color of each point, defined by spherical harmonics (SH) coefficients. The density α_i is computed as the product of 2D Gaussians and a learnable point-wise opacity.



(I) Pinhole-based model (II) Lens-based model

Figure 3. **The difference between the pinhole-based model** (I) **and the lens-based model** (II)**.** For (I), light emitted from spatial point O directly hits the point O' of the image plane. For (II), we show the case that light emitted from Q converges on the point Q' of the image plane (**In-Focus**), and the case that light emitted from P converges on the point P' and continues to scatter onto the image plane forming a circle of confusion (**Out-of-Focus**).

4. Method

We propose *DoF-Gaussian*, a controllable DoF approach for 3D-GS. The overall pipeline is illustrated in Fig. 2. In Section 4.1, we first develop a lens-based imaging model based on geometric optics principles to control DoF effects. Next, we employ the per-scene adjustment of depth priors to guide the correct scene geometry, as described in Section 4.2. Finally, in Section 4.3, we apply defocus-to-focus adaptation to minimize the inconsistencies in the CoC and enhance the defocus deblurring performance.

4.1. Lens-based Depth-of-Field Model

The physical principles underlying imaging and DoF have been extensively studied in the field of geometric optics [10, 32]. In an idealized optical system, the light emitted from a spatial point O is projected onto a corresponding point O' in the image plane following the principles of pinhole imaging, as illustrated in Fig. 3 (I). However, real-world cameras operate based on the lens model, as shown in Fig. 3 (II). We show the two cases of in-focus and out-of-focus imaging, respectively. A spatial point P, located at a distance d from the lens, is projected on the imaging plane as a circular region referred to as the circle of confusion (CoC). The diameter of this region, $r(d)$, can be determined by the aperture parameter $\mathcal{A}$ and focus distance $\mathcal{F}$, according to the following equation:

$$r(d) = \mathcal{A}\left|\frac{1}{\mathcal{F}} - \frac{1}{d}\right|. \tag{5}$$

The focus distance $\mathcal{F}$ primarily controls the depth position of the focal plane of the image, *i.e.*, the depth of the sharp area, while the aperture parameter $\mathcal{A}$ determines the extent of the bokeh effect. For the point Q located at the focus distance $d = \mathcal{F}$, the emitted rays directly converge through the lens to the corresponding point Q' in the image plane, resulting in the absence of a circle of confusion (CoC). This also implies that $r(\mathcal{F}) = 0$.

Algorithm 1 Lens-based imaging process

Input: Rasterization rendering C, aperture parameter $\mathcal{A}$, focus distance $\mathcal{F}$, depth map D, gamma value γ, confuse function $Func$
Output: Defocus simulated result C^*

```
R ← A|1/F − 1/D|    C ← (C)^γ
Φ = [0], C* = [0]
for pixel i ← Traverse(C) do
    for pixel j ← TraverseNeighbor(c_i, r_i) do
        λ_ij ← Func(d_i, |i − j|)
        Φ_j ← Φ_j + λ_ij
        c*_j ← c*_j + λ_ij · c_i
    end for
end for
C* ← (C*/Φ)^(1/γ)
```

We set focus distance $\mathcal{F}$ and aperture parameter $\mathcal{A}$ as learnable for each input image, and these parameters are continuously updated with the optimization of 3D-GS. By modeling the lens camera, the output C^*, which includes bokeh blur effects, can be derived from the 3D-GS rasterizer rendering result C using our CUDA-based Algorithm 1.

Ideally, the confuse function is represented by the indicator function $\mathbb{I}(r(d) > l)$. To achieve a smooth and natural DoF effect, we substitute it with a differentiable function, as proposed in Busam *et al.* [3]:

$$Func(d, l) = \frac{1}{2} + \frac{1}{2}\tanh\Big(\alpha\big(r(d) - l\big)\Big), \tag{6}$$

where α defines the smoothness of confuse transition and l represents the distance between two pixels.

We supervise the training by computing the reconstruction loss between the output image C^* and the shallow DoF input image I, where the loss function is a combination of the $\mathcal{L}_1$ and a D-SSIM term, with λ set to 0.2:

$$\mathcal{L}_{rec} = (1 - \lambda)\mathcal{L}_1(I, C^*) + \lambda\mathcal{L}_{D-SSIM}(I, C^*). \tag{7}$$

For inference, we set the aperture size $\mathcal{A}$ to 0 so that we can render fully clear images for novel view synthesis.

4.2. Per-Scene Adjustment of Depth Priors

Unlike previous work that assumes input images are all-in-focus and fully clear, it becomes challenging to accurately reconstruct scene geometry and render depth maps for bokeh blurring input images. However, rendering images with varying depths of field relies on precise depth information. To address this issue, we propose using depth prior as useful guidance for the accurate reconstruction of scene geometry. However, due to the presence of bokeh blur in shallow DoF inputs, directly employing a monocular depth network to predict depth maps D_{pred} as pseudo

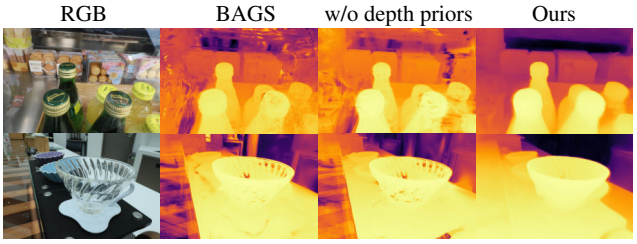


Figure 4. **Effects of the depth priors.** The per-scene adjustment of depth priors enhance the geometric structure and yield a more accurate depth map than that estimated by BAGS [34] and a variant of our method without depth priors.

ground truth to regularize the Gaussians rendered depth map, as done in [5, 21], does not yield satisfactory results. Inspired by [53], we propose adapting per-scene depth priors through fine-tuning a monocular depth network based on the scene's sparse reconstruction. Specifically, we utilize COLMAP [40, 41] to obtain a sparse Structure-from-Motion (SfM) point cloud and create a set of 3D Gaussians. Consequently, we can derive per-view sparse depth maps D_{sparse} by projecting the 3D point clouds after multi-view stereo processing. Although the resulting depth map is sparse, it is robust and can serve as a supervisory signal for training the per-scene depth priors. To address the scale ambiguity inherent in acquired depth maps, we apply a silog loss [9] to fine-tune the depth network [22, 39], expressed as:

$$\mathcal{L}_{silog} = \frac{1}{2M}\sum_{i=1}^{M}(\log(e^s D_{pred}) - \log(D_{sparse}))^2, \quad (8)$$

where e^s is the scale factor and M is the number of pixels in the image used for calculation. We apply the rasterization technique from Rade-GS [59], which enables the robust rendering of depth maps D for 3D-GS scenes. We predict the depth maps D_{pred} from multi-view input images using the fine-tuned depth network and use them as depth priors to regularize D. The depth loss can be described as follows:

$$\mathcal{L}_{depth} = ||D - D_{pred}||_2. \quad (9)$$

As illustrated in Fig. 4, it is evident that the depth priors significantly enhance the geometric structure, resulting in a more accurate depth map compared to both BAGS [34] and the variant of our method without depth priors. In Section 5.3, we further conduct an ablation study comparing several other depth supervision strategies to demonstrate the effectiveness of our approach.

4.3. Defocus-to-Focus Adaptation

The CoC effect achieved under ideal optical imaging conditions, as described in Algorithm 1, inevitably differs from the real CoC effect produced by an actual DSLR camera. This discrepancy hinders our ability to accurately model the out-of-focus blur in shallow DoF input images.

To achieve sharper scenes, we propose an adaptation process that allows transitions from defocus to focus. Specifically, modeling the defocus effects with bokeh blur facilitates learning accurate lens parameters. Consequently, we place greater emphasis on capturing the sharp areas in training images once the lens parameters $\mathcal{A}$ and $\mathcal{F}$ have been optimized and converged. This approach is effective because, when the focus distance is determined, the sharp regions in the scene become approximately identifiable. After t optimization iterations, we reweight the $\mathcal{L}_{rec}$ and $\mathcal{L}_{depth}$ loss by employing a step-like function:

$$\Psi(x) = \begin{cases} 1 & , \text{iterations} < t \\ 1/(1+e^{-a\cdot(x-b)}) & , \text{iterations} \geq t \end{cases} \quad (10)$$

where a and b are hyperparameters, and $x = |\frac{1}{\mathcal{F}} - \frac{1}{d}|$.

In addition, the aperture size $\mathcal{A}$ is generally large after the convergence, which can cause blurring of pixels that should be in focus. Hence, we also reweight the aperture size $\mathcal{A}$ on a pixel-wise basis:

$$\mathcal{A}' = \mathcal{A} \cdot \Psi. \quad (11)$$

Full Objective. We derive the final loss terms by incorporating the reconstruction loss from Eqn. 7, the depth loss from Eqn. 9, and an additional normal consistency loss as proposed by Huang *et al*. [11]. The normal consistency loss ensures that the Gaussian splats align with the surface by measuring the consistency between the normal directions computed from the Gaussian and the depth map:

$$\mathcal{L}_{normal} = \sum_i w_i(1 - n_i^{\mathsf{T}}\hat{n}_i), \quad (12)$$

where $\hat{n}$ represents the surface normal direction obtained by applying finite differences on the depth map, i indexes the intersected splats along the ray and w denotes the blending weight of the intersection point.

Our final training loss $\mathcal{L}$ is:

$$\mathcal{L} = \Psi \odot (\mathcal{L}_{rec} + w_d\mathcal{L}_{depth}) + w_n\mathcal{L}_{normal}, \quad (13)$$

where $\odot$ is Hadamard product.

5. Experiments

Implementation Details. In our experiments, we set the smoothness of confuse transition $\alpha = 4$. In addition, we set $a = 15$, $b = 0.3$, and $t = 10000$ for the defocus-to-focus adaptation. Our method is built upon Mip-Splatting [58], and our optimization strategy and hyperparameter settings remain consistent with it. We train each scene for 30000 iterations and set loss weights $w_d = 0.01$ and $w_n = 0.05$.

Table 1. **Quantitative comparisons on the defocus blur dataset of Deblur NeRF [26].** We report the PSNR, SSIM, and LPIPS metrics and color each cell as best and second best. Our method outperforms other existing approaches across most scenes.

Method	Deblur-NeRF [26]			DoF-NeRF [55]			DP-NeRF [19]			BAGS [34]			Deblurring 3DGS [18]			**Ours**		
	PSNR	SSIM	LPIPS	PSNR	SSIM	LPIPS	PSNR	SSIM	LPIPS	PSNR	SSIM	LPIPS	PSNR	SSIM	LPIPS	PSNR	SSIM	LPIPS
Cake	26.27	0.780	0.128	24.53	0.731	0.206	26.16	0.778	0.127	27.21	0.818	0.108	26.88	0.803	0.116	26.83	0.808	0.101
Caps	23.87	0.713	0.161	22.67	0.636	0.261	23.95	0.712	0.143	24.16	0.725	0.159	24.50	0.742	0.149	24.62	0.749	0.147
Cisco	20.83	0.727	0.087	20.64	0.724	0.107	20.73	0.726	0.084	20.79	0.743	0.070	20.83	0.732	0.079	21.00	0.744	0.069
Coral	19.85	0.600	0.121	19.83	0.570	0.240	20.11	0.611	0.118	20.53	0.628	0.111	19.78	0.608	0.131	20.37	0.630	0.109
Cupcake	22.26	0.722	0.116	21.89	0.706	0.143	22.80	0.741	0.096	22.93	0.762	0.080	22.11	0.734	0.099	22.97	0.757	0.079
Cups	26.21	0.799	0.127	25.26	0.765	0.202	26.75	0.814	0.104	26.27	0.823	0.104	26.28	0.824	0.103	26.01	0.817	0.100
Daisy	23.52	0.687	0.121	23.22	0.658	0.194	23.79	0.697	0.108	23.74	0.746	0.062	23.54	0.731	0.095	23.93	0.735	0.071
Sausage	18.01	0.500	0.180	17.86	0.488	0.280	18.35	0.544	0.147	18.76	0.574	0.110	18.99	0.570	0.141	19.11	0.576	0.119
Seal	26.04	0.777	0.105	24.85	0.687	0.143	25.95	0.778	0.103	26.52	0.812	0.090	26.18	0.817	0.098	26.57	0.825	0.087
Tools	27.81	0.895	0.061	26.21	0.854	0.128	28.07	0.898	0.054	28.60	0.913	0.046	27.96	0.910	0.058	28.29	0.913	0.051

Table 2. **Quantitative comparisons on our synthetic dataset.**

Method	PSNR↑	SSIM↑	LPIPS↓	$\delta_{\mathcal{A}}\downarrow$	$\delta_{\mathcal{F}}\downarrow$
DoF-NeRF [55]	25.59	0.788	0.207	0.196	0.256
Ours	**28.70**	**0.864**	**0.095**	**0.126**	**0.079**

Table 3. **Comparisons in all-in-focus settings.** We compare our method with Mip-Splatting on all-in-focus datasets to validate the effectiveness of our model under general input conditions.

Method	PSNR↑	SSIM↑	LPIPS↓
Mip-Splatting [58]	27.05	0.893	**0.115**
Ours	**27.81**	**0.902**	0.117

Baselines and Evaluation Metrics. We compare our method with state-of-the-art deblurring techniques [18, 26, 34] and a depth-of-field method [55]. To quantitatively evaluate the quality of novel view images, we adopt widely used metrics such as PSNR, SSIM [52], and LPIPS [60]. We will compare with Wang *et al*. [51] once they release the code or provide the full experimental results in their paper.

Datasets. Following [18, 26, 34], we evaluate our model on the Deblur-NeRF dataset [26]. Subsequently, we conduct further evaluations on the Real Forward-facing dataset [28] and T&T_DB dataset [16]. The effectiveness of our method is validated on these datasets for both shallow DoF inputs and normal inputs. Furthermore, we also introduce a dataset in Section 5.1 to evaluate our model more comprehensively.

5.1. A Synthetic Dataset

We introduce a synthetic dataset, as illustrated in Fig. 6, based on depth estimation [2] and a single-image DoF rendering method [35] for each image in the Real Forward Facing dataset [28] and T&T_DB dataset [16]. This allows us to evaluate the refocusing ability of our model and to determine whether it accurately learns the correct aperture size and focus distance. In particular, by feeding the DoF rendering method [35] the known aperture size and focus distance, we can convert the all-in-focus images into shallow DoF images. Unlike existing defocus blur datasets proposed by Ma *et al*. [26] and Wu *et al*. [55], the test set in our dataset comprises images with shallow DoF. With known lens parameters, our model fits this shallow DoF effect to quantitatively measure the refocusing ability. Meanwhile, we establish the known lens parameters beforehand as ground truth, allowing us to calculate the errors $\delta_{\mathcal{A}}$ and $\delta_{\mathcal{F}}$ between the learned parameters and the ground truth. More details can be found in the supplementary material.

5.2. Comparisons

Quantitative Comparisons. We quantitatively compare *DoF-Gaussian* against other state-of-the-art methods using a real-world defocus blur dataset [26]. As shown in Table 1, our method outperforms other state-of-the-art baselines. These results suggest that our method achieves superior defocus deblurring performance and produces higher-quality novel view synthesis images.

Qualitative Comparisons. Visual qualitative comparisons are presented in Fig. 5. Our method surpasses all other methods in terms of image fidelity, generating novel view images that are more faithful to the ground truth images and exhibit less blur. Specifically, the bottle caps and shelves are the sharpest in our rendered images.

All-in-focus Inputs. To further validate the effectiveness of our method in the all-in-focus setting, we design an experiment using wide DoF images as inputs. We conduct a comparison between our method and Mip-Splatting [58] on two all-in-focus datasets, the Real Forward Facing dataset [28] and the T&T_DB dataset [16]. As shown in Table 3, our method achieves comparable or even better performance to Mip-Splatting on the average metrics across the datasets. This demonstrates that our lens-based model not only excels with shallow DoF inputs but also achieves good

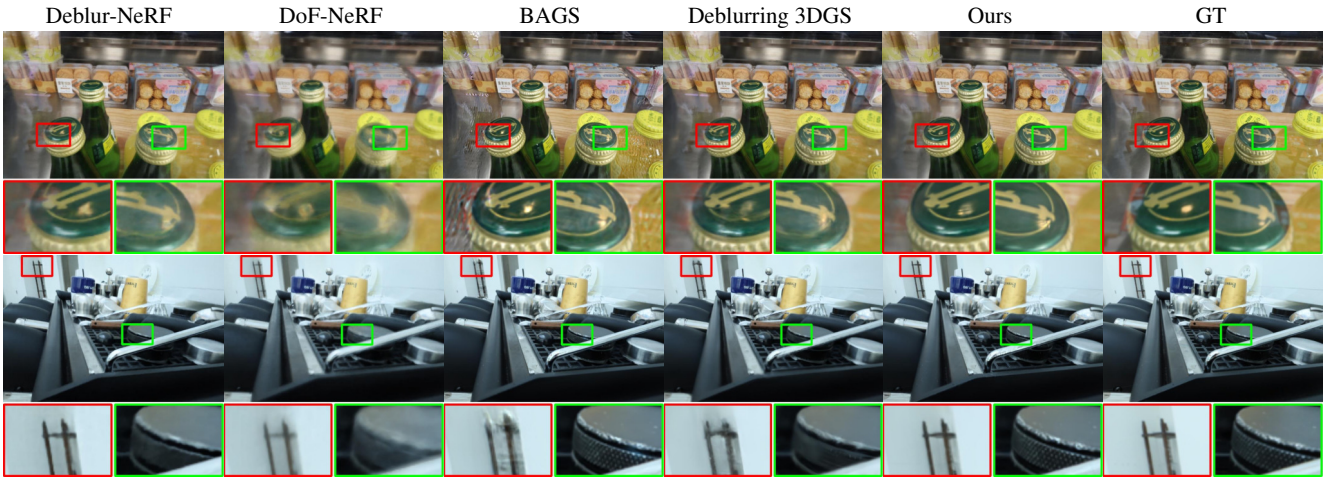


Figure 5. **Qualitative comparisons against all baselines.** Compared to other state-of-the-art methods, our method represents sharper scenes and generates novel view images with less blur.

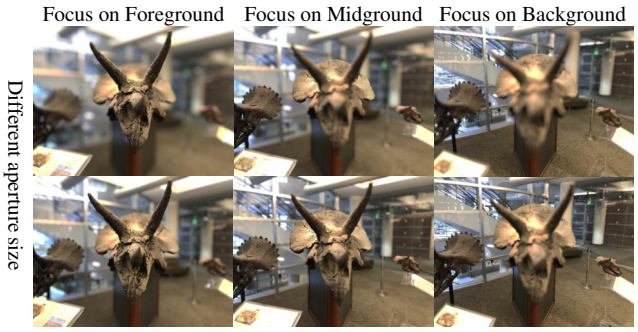


Figure 6. **Examples in our dataset.** For each scene, we apply a DoF rendering method [35] to convert all-in-focus images to shallow DoF images with varying focus locations and aperture sizes. This dataset is used to evaluate the model's refocusing ability.

results with general input images.

Our Synthetic Dataset. As shown in Table 2, our method achieves superior refocusing ability through controlled DoF rendering, producing rendered images with the highest quality on our synthetic dataset. Furthermore, our method learns more precise aperture size and focus distance than DoF-NeRF [55]. Since Deblur-NeRF [26], BAGS [34], and Deblurring 3DGS [18] do not incorporate a lens model, they cannot generate novel view images with a specific DoF. As a result, these methods cannot be evaluated on this dataset due to their lack of refocusing capability.

5.3. Ablation Study

As shown in Table 4 and Fig. 8, we conduct ablation studies to evaluate the effectiveness of our designs. The comparison between #1 and #2 indicates that the lens-based imaging model has a significant impact. This lens model not only facilitates defocus deblurring but also provides the ability to control the depth of field. Next, the defocus-to-focus adaptation (#3) and the per-scene adjustment of depth priors (#4) further enhance the scene geometry and improve the deblurring of details. Ultimately, the combination of all components yields the highest performance gain, leading to a 2.66 dB increase in PSNR over the baseline model.

Table 4. **Ablation study on each component of our method.**

	Lens	Depth	Defocus-to-focus	PSNR↑	SSIM↑	LPIPS↓
#1	✗	✗	✗	21.31	0.636	0.239
#2	✓	✗	✗	23.05	0.728	0.109
#3	✓	✗	✓	23.59	0.742	0.104
#4	✓	✓	✗	23.42	0.738	0.098
#5	✓	✓	✓	**23.97**	**0.756**	**0.093**

Table 5. **Ablation study on different depth priors.**

Method	PSNR↑	SSIM↑	LPIPS↓
No fine-tuned depth	23.53	0.739	0.114
Sparse depth	23.44	0.737	0.113
Ours	**23.97**	**0.756**	**0.093**

In addition, we conduct an ablation study on different depth supervision methods to demonstrate the superiority of our per-scene adjustment of depth priors, as shown in Table 5. Direct supervision using depth maps predicted by the depth network without scene-specific fine-tuning hinders the reconstruction quality of 3D-GS. Likewise, supervision based on sparse depth maps projected from a 3D point cloud fails to yield optimal results. We show visualizations of depth maps rendered by different depth supervision strategies in the supplementary material.

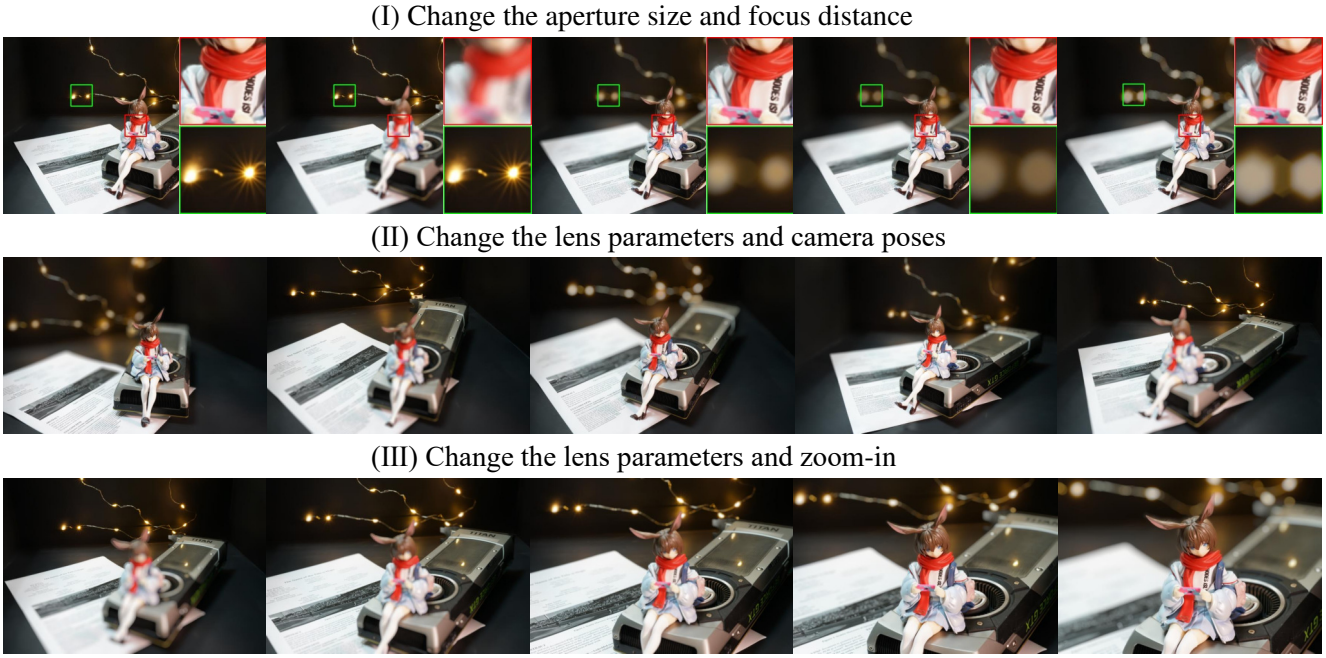


Figure 7. **Applications of our controllable DoF effects.** Users can create their own cinematic moments by combining changes in aperture size, focus distance, camera poses, and zoom.

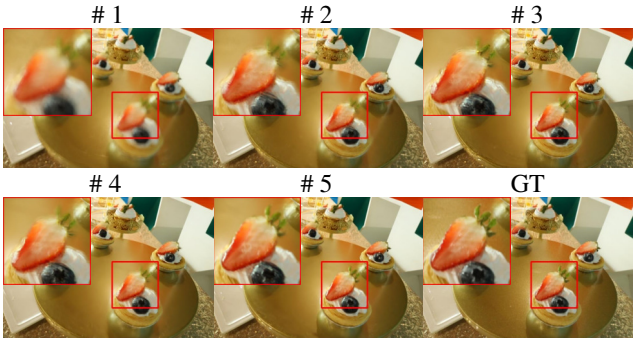


Figure 8. **The visualizations of the ablation study.** Qualitative comparisons show that the full model (#5) yields the best result and the reconstructed strawberry is the sharpest.

5.4. Interactive Applications

Thanks to the lens-based model, we can not only represent 3D scenes from shallow DoF inputs but also render interactive DoF effects (see Fig. 7). Various cinematography styles can be achieved through different camera configurations, and users can input custom datasets, whether all-in-focus or bokeh images. By controlling aperture size and focus distance, we can render novel view images with different depths of field. In addition, we can modify the shape of the CoC, such as changing it from a circle to a hexagon. Users can also adjust the depth of field while moving the camera or zooming to create cinematic effects. Unlike [55], our GS framework significantly enhances the training and rendering efficiency, allowing for faster and more seamless interactions. The details on processing time will be discussed in the supplementary material.

6. Conclusion

While 3D Gaussian Splatting methods have achieved impressive performance on a wide range of reconstruction tasks, the applications on controllable DoF effects remain challenging and understudied. In this paper, we propose DoF-Gaussian, a controllable DoF method for 3D Gaussian Splatting. Specifically, we develop a lens-based model rather than pinhole imaging to overcome the limitations imposed by shallow DoF inputs for 3D-GS. Furthermore, we propose the per-scene depth adjustment and a defocus-to-focus adaptation to guarantee the performance of defocus deblurring. We also introduce a synthetic dataset for a more comprehensive evaluation. Thanks to the imaging principles, our method supports various interactive applications.

Future work. Inspired by our experiments, we found that modeling real-world physical imaging principles can enable our method not only to handle shallow DoF inputs but also to perform effectively on general inputs. This insight motivates us to explore a combination of non-ideal conditions, such as shallow DoF inputs, sparse views, and varied lighting environments, which are more aligned with casual photography in our daily lives.

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